Sparse mixtures of linear transforms

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This work is a preliminary update describing a method we are still developing. As such, many of the results w present here are incomplete. We present them in the hopes that it inspires follow-up work and improvement from the external research community.

Upon publishing this update, we became aware of contemporaneous work on a highly-related architecture ("Mixture of Decoders," Oldfield et al.). We have updated the post to describe the similarities / differences between our methods in the Related Work section. We recommend that anyone interested in this work read theirs as well!

In our recent work, we trained transcoders – sparse, extra-wide MLPs – as more interpretable replacements for model's original MLP layers. We used the transcoder neurons ("features") as a basis for understanding model computation. We described computations using "attribution graphs," which depict the causal interactions betw features that give rise to the model's outputs.

While this approach has proved very useful, we believe that transcoder features have key limitations. Transcode "shatter" model computation into many extremely granular pieces. This can lead them to represent computatic a different, less efficient way from the underlying model, and can introduce pathologies like feature splitting an absorption.

To take one specific example, we found that transcoders decomposed addition circuits using "lookup table" features that represented individual, highly specific computations, like "6 + 9 = 5 (mod 10)". Other work sugge that transformers embed their representations of numbers in a geometric structure such that simple transforma can be used to compute arithmetic operations. While the lookup table description of addition is accurate in sor sense, it misses key structure in the model, and is highly inefficient. This inefficiency is a serious issue in practice order to train transcoders that are sufficiently large to capture all the computation in a model, we would likely r an astronomical number of features, using a transcoder with many more parameters than in the underlying mo

In this preliminary update, we provide a description of a method we have been working on that is intended to address some of these concerns. Our approach is to replace MLPs with a sparse *mixture of linear transforms* (M Unlike a transcoder, a MOLT does *not* learn sparsely active *features* that are embedded along a vector direction model's activation space. Instead, it learns sparsely active *transforms*, which apply a linear transformation to th residual stream to give their contribution to the MLP output. Unlike transcoder features, which double as both computational and representational units, MOLT transforms are purely computational objects that "bridge" representations between layers. As such, they are not intended to be studied on their own, but in combination another representation decomposition algorithm like SAEs.

So far, we have found that MOLTs are a more compute-efficient and mechanistically faithful way of replacing M than transcoders are. We find that the conditions in which MOLT transforms are active are similarly interpretabl transcoder features. Our preliminary experiments suggest that these transforms can be used to understand hov features in one layer are transformed into features in a subsequent layer. MOLT transforms can be incorporatec an attribution graph by annotating graph edges with the set of transforms that "carried" the edge. We also sus (but have not yet shown) that *compositions* of MOLT transforms can be used to understand compositional struc in model representations.

Implementation

A MOLT is parameterized as follows:

$$f(x) = \sum_t [\phi(\mathbf{e_t} \cdot \mathbf{x} - b_t) * (U_t V_t \mathbf{x})]$$

where

- $\mathbf{x}$ is the input residual stream activations
- t indexes the transforms
- $\mathbf{e_t}$ and b_t are the encoder vector and bias for transform t, which determine whether the transform is ac and how active it is. ϕ is a nonlinearity like ReLU or JumpReLU
- U_t and V_t are matrices of dimensions $d_{model} \times k_t$ and $k_t \times d_{model}$, respectively. Thus, $U_t V_t$ is a rank-k matrix. Different transforms may have different ranks (we treat the distribution of ranks as a hyperparan to be optimized).

A MOLT is trained to mimic the output of a model MLP layer, just like a transcoder is. We apply a sparsity penal (e.g. L1 or tanh) to the activations of the transforms, scaled by the Frobenius norm of the transform matrix (i.e. penalize $\|U_t V_t\|_F \cdot \phi(\mathbf{e_t} \cdot \mathbf{x} - b_t)$). This is similar to how one applies a sparsity penalty to the activations of transcoder features, scaled by the norm of their decoder vector.

Related Work

The *mixture of decoders (MxD)* architecture (Oldfield et al.) is quite similar to ours – it replaces MLP layers with a sparse mixture of linear transforms. There is one key implementation difference between MxDs and MOLTs. Us sparse mixture of many independently learned, full-rank transforms is intractable because it would require too parameters. MOLTs get around this issue by making the linear transforms low-rank (with a distribution of differe ranks; see below). MxDs allow the transforms to be full-rank, but parameterize them in a way that shares param across different transforms.

Specifically, (translating the MxD paper notation into ours), each $U_t V_t$ transform has full rank, but the hth colu the U_t matrix for the tth transform is parameterized as an elementwise product of a vector $\mathbf{d}_h * \mathbf{c}_t$, where the vectors are shared across transforms and the $\mathbf{c}$ vectors are shared across column indices. We are not yet confic the pros and cons of the MxD vs. MOLT strategies. We suspect that our low-rank transform parameterization incentivizes transforms that perform more interpretable, tightly-scoped computational roles, but more work is needed to compare the two approaches.

We also note that our intended use-cases for MOLTs (described later in the post) are somewhat different than t focus of the MxD paper – in particular, we are especially interested in interpreting MOLTs as implementing (potentially compositional) transformations between residual stream features (e.g. from SAEs), and integrating t into attribution graphs, treating MOLT transforms as a kind of MLP analog to attention heads.

MOLTs also bear some resemblance to skip transcoders, which augment transcoders with a linear transform; ho they differ in several ways:

- MOLT transforms are only conditionally active, and penalized to be active infrequently
- MOLT transforms are not full-rank
- MOLTs have multiple linear transforms, rather than just one
- MOLTs do not have any "regular" transcoder features

Intuition

What's the intuition for why sparsely active transforms might be a good way to represent computation? We off few perspectives.

- First, we note that MOLTs have a rough correspondence to how MLPs do computation. Consider an ML layer that uses ReLU neurons. If you condition on the set of active neurons, the MLP is just applying a li transformation to its input. Patterns of active neurons thus correspond to linear transformations, with r equal to the number of neurons in the pattern. These can be thought of as loosely analogous to a MOL transform.

- Sparsely active linear transforms can capture some ways in which the model leverages geometry of the residual stream to perform computations. As noted above, transcoders require "lookup table features" t select for specific pairs of inputs in order to perform addition. By contrast, a MOLT can implement addit more efficiently. Suppose the digits 0-9 are represented by the model on a circle. A rank-2 "plus 3" transform, could represent this by performing a rotation 8/10ths of the way around the circle, and coulc transform a "1" feature into a "say 4" feature in the next layer, a "2" feature into a "say 5" feature, and sc The "plus 3" transform's *encoder* would check for textual cues in the context like "+3" or "add three."

- Another perspective is that MOLTs are a way to compositionally parameterize features. If we have unco 1 million features at the residual stream input to layer L, and uncovered 1 million transforms in layer L M we have effectively described 1 *trillion* "pseudofeatures" in layer L+1, corresponding to pairs of features transforms (a pseudofeature is "active" when its corresponding feature and transform are active, and its "direction" is the result of applying the transform matrix to the feature decoder). Moreover, if we consic features propagated through chains of multiple transforms, the number of "pseudofeatures" we capture grows exponentially in the chain length.

Results

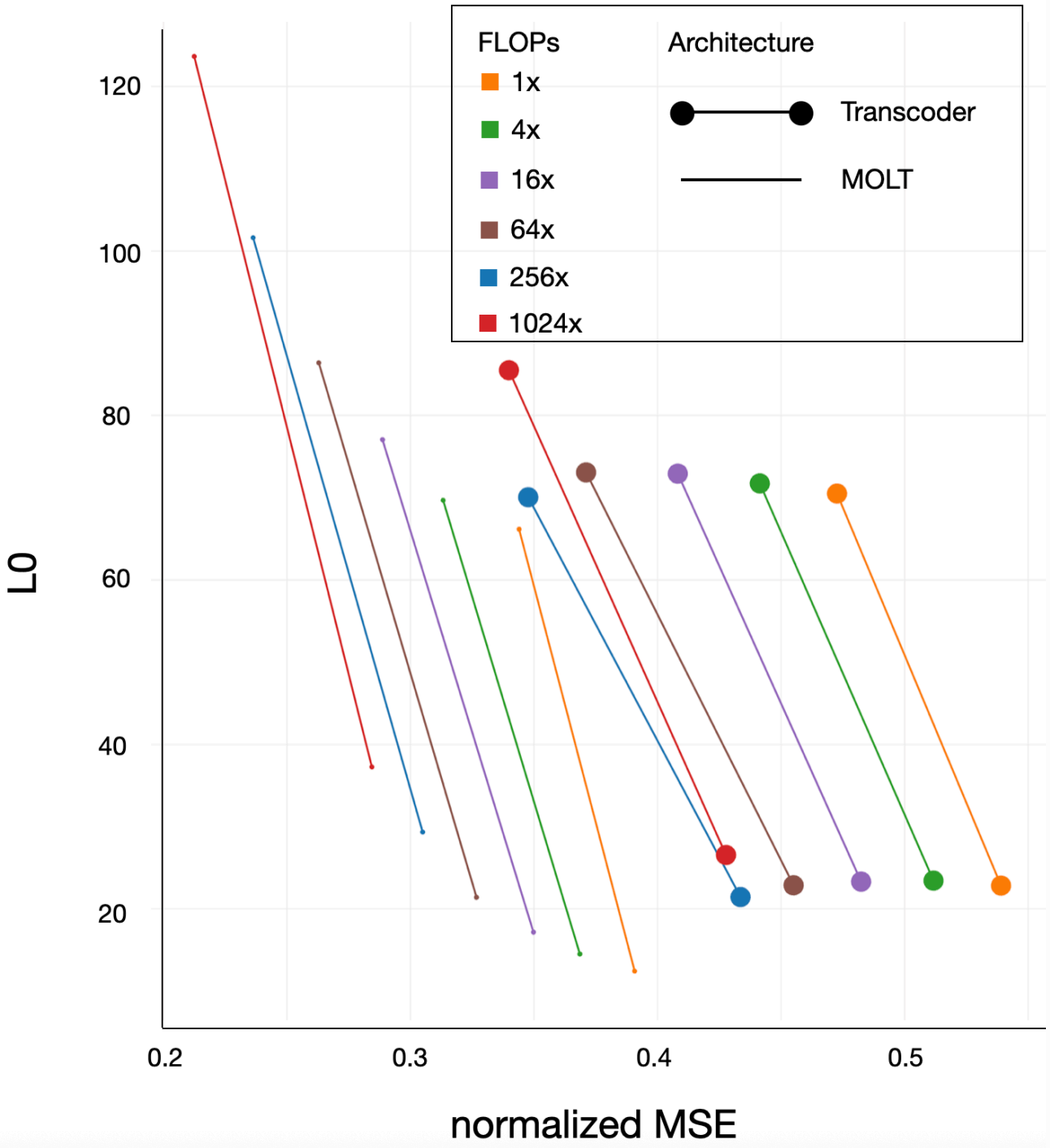
OPTIMAL ALLOCATION OF RANKS

An important hyperparameter in training a MOLT is the allocation of ranks to transforms. This is a high-dimens hyperparameter space, and we have not explored it fully. However, when training MOLTs on Claude 3.5 Haiku, have obtained our best performance (in terms of the MSE/L0 pareto frontier) using a distribution of ranks, vary from 32 to 512. Concretely, we use a collection of N transforms of rank 512, 2N of rank 256, 4N of rank 128, 8N rank 64, and 16N of rank 32. To increase the scale of runs we vary N, but keep the proportions the same. We ha found that using transforms of variable ranks outperforms using transforms of all the same rank, controlling for total number of parameters.

ML PERFORMANCE

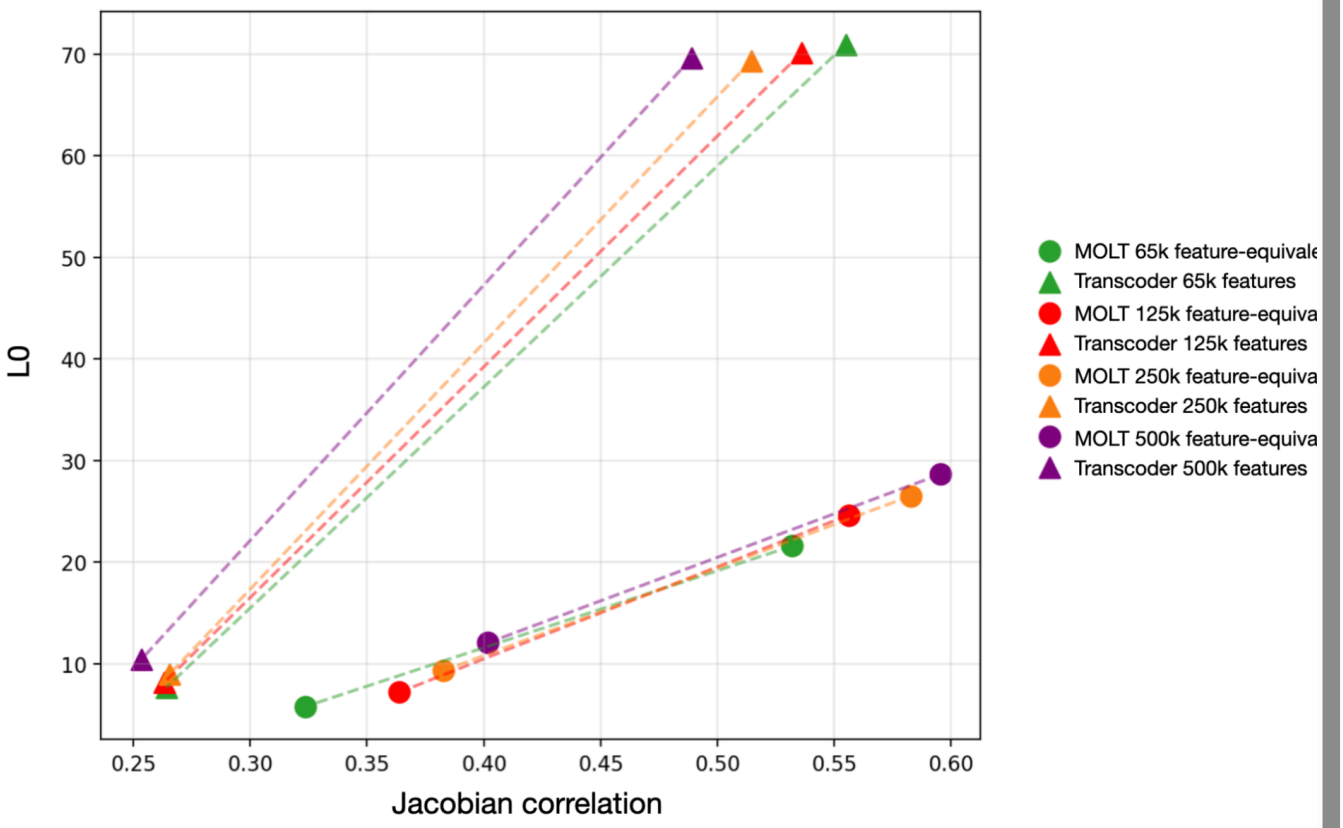
We trained MOLTs (using the rank allocation given above) and transcoders on the middle layer of Claude 3.5 Ha varying the amount of compute used in the run. We scaled the number of training steps proportionally to the number of features, and matched the number of parameters between transcoder and MOLT runs. Thus each 4x increase in FLOPs reflects a 2x increase in both number of parameters and training steps. The largest ("1024x FL transcoder runs contain approximately 10 million features.

We find that at a given L0, the reconstruction error (MSE) is significantly lower for MOLTs than transcoders, controlling for the number of parameters. The smallest MOLT runs here Pareto-dominate transcoder runs that u 1024x as many FLOPs. Moreover, transcoder performance appears to be saturating at the higher compute scale (though it is possible this flattening is simply due to poor ML tuning), while we observe no such saturation for b runs.



We also evaluated the mechanistic faithfulness of MOLTs compared to transcoders – that is, the degree to whic MOLT (or transcoder) responds to input perturbations in the same way as the underlying MLP layer. In the limit infinitesimal perturbation sizes, faithfulness can be computed by comparing the Jacobians of the replacement l to the underlying layer on a given datapoint, and averaging over datapoints. We find that MOLTs have a substa higher Jacobian correlation (cosine similarity of the flattened Jacobian matrices) than transcoders do, at the san moreover, the faithfulness of transcoders appears to deteriorate with scale, whereas that of MOLTs is more stab The greater faithfulness makes sense, given that the Jacobians of transcoders are constrained to be low-rank (r upper-bounded by the L0), whereas MOLT Jacobians can have rank much higher than their L0.

Note that the results below are from a different model than those above (the 18-layer model used in our circuit tracing paper, rather than Claude 3.5 Haiku); we have not yet performed this Jacobian analysis on Haiku. MOLT with a given number of "feature-equivalents" have the same number of parameters as a transcoder run with tha many features.



TRANSFORM INTERPRETABILITY

Transforms are characterized by two properties:

- Under what conditions are they active (and how active)
- What function do they perform when active

To understand the first part, we can use the same visualization strategy we use for SAE and transcoder features highlighting dataset examples that activate the transform. When we do so, we find that transforms appear qualitatively similar to features – we see transforms that select for token-level information in earlier layers, and transforms that select for more abstract contextual information in middle and later layers.

We also observe that higher-rank transforms skew higher-density (i.e. are active more often).. However, the hig density, higher-rank transform conditions still appear comparably interpretable to lower-density transforms (an transcoder features). For instance, we observed a high-rank transform that activates on period tokens, and anot that activates on text written in Spanish.

Interpreting the function of transforms is more difficult. Our initial strategy was to train SAEs on the residual str prior to and immediately after the MOLT layer, and identify feature-feature pairs that most strongly interact via given transform ($\mathbf{e}_t UV \mathbf{d}_s$, where $\mathbf{e}_t$ is the target feature encoder and $\mathbf{d}_s$ is the source feature encoder). Howe we found this information difficult to interpret, presumably due to the same problem of interference weights th makes raw interaction strengths between transcoder features difficult to interpret.

We have had more success interpreting transforms in the context of attribution graphs, as described below.

INTEGRATING MOLTS INTO ATTRIBUTION GRAPHS

METHOD

In our recent paper, we constructed attribution graphs built on top of transcoder features. Edges between featu were computed by determining the influence that a source feature (via its decoder direction) exerted on a targe feature's encoder direction (either directly, via residual connections, or via attention heads).

MOLT transforms have no fixed decoder direction that they write out to, so the same attribution graph strategy cannot be applied to them. However, we have had preliminary success with another attribution graph strategy

- Train SAEs on each residual stream layer of the model (or a cross-layer variant, like weakly causal crossc We find graphs benefit from cross-layer dictionaries, but we will explain the per-layer SAE case for simp
- For each pair of source and target features, decompose the attribution between the features into a sum terms of two kinds:
    - Terms mediated by MOLT transforms (attributions via MLP layers). For source feature i with dec $\mathbf{d}_i$ and activation a_i, target feature j with encoder $\mathbf{e}_j$ and activation a_j, and transform UV wit activation a_t, the term will be $\mathbf{e}_j UV \mathbf{d}_i \cdot (a_i a_t)$.
    - Terms mediated by attention heads (attributions via attention layers). For source feature i with decoder $\mathbf{d}_i$ and activation a_i at token position q, target feature j with encoder $\mathbf{e}_j$ and activatio at token position k, and attention head with an OV matrix with attention pattern $a_h(q,k)$ bet the token positions, the term will be $\mathbf{e}_j OV \mathbf{d}_i \cdot (a_i a_h(q,k))$.
- Annotate each edge in the attribution graph with the list of MOLT transforms and attention heads that strongly mediated that edge. Hovering over MOLT transforms surfaces their "viz" – a panel displaying t top-activating dataset examples. (Clicking on attention heads reveals an interface that displays their "Q attributions," to be described in another update, TK).
- Note that MOLT transforms themselves have input edges, computed the same way as above, but using transform's encoder vector as the attribution target

QUALITATIVE FINDINGS

Using such attribution graphs, we have uncovered instances of MOLT transforms performing interpretable computations, such as:

- In the prompt The Spanish word for hot is "calor", we observed that "say a word beginning with 'cal'" features received input from "hot" features via a transform that is active in Spanish-language-related

contexts.

    - Interestingly, we also see some inputs from “Spanish” features via transforms that are active in “ related contexts!

- In the prompt 3 + 5 = 8, we observed a “say 8” feature receiving input from a “plus 3” feature via a tran active in contexts in which the number “5” recently appeared

However, we also see transforms playing roles that are less clearly interpretable. For instance, many edges appe be mediated by transforms that select for key words like “is” or “Assistant.” We also see transforms playing role appear redundant with the features they carry, such as a “Paris” feature receiving input from a “France” feature “France” transform.

Conclusion

We believe MOLTs are a promising alternative to transcoders and may be able to capture MLP computation in a parameter-efficient way that more faithfully reflects the computations performed by the underlying model. We suspect that a MOLT-like solution will be necessary to capture all the variance of frontier model MLP layers at a reasonable computational cost. We see signs of life that MOLT transforms can perform interpretable computati “transforming” input features into output features.

One direction we are excited about, but have not explored yet, is using MOLT transforms to understand compositional representations that are not captured by our SAEs. In particular, the reconstruction error of our at each layer can be rewritten as a sum of terms corresponding to (feature, transform) pairs from the *previous* la These might correspond to concepts that are too rare to be captured by our finite-size SAEs, but are built out o *composing* a relatively common feature with a relatively common transform. In an attribution graph, decompos SAE errors into (feature, transform) pairs in this fashion would manifest as having some graph edges that are mediated by *chains* of transforms in consecutive layers.

More work needs to be done to conclude that MOLTs are strictly preferable to transcoders. Attribution graphs include transform information are somewhat more unwieldy than transcoder-based attribution graphs, and not transform-mediated computations are clearly interpretable. Future work on scaling and improving MOLTs, and associated attribution graph logic and UI, may address some of these issues.